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SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

Master of Science in Information Technology

MIT8102: Advanced Distributed Systems

TERM PAPER

DISTRIBUTED GRID COMPUTING:

Architectures, Security, Middleware, and Emerging Paradigms

Prepared by:

Kathembo Tsongo Dieudonne **112721**

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Introduction

Background

The origins of Distributed Grid Computing lie in the metacomputing experiments of the early 1990s. The I-WAY experiment of 1995 linked supercomputers at seventeen North American sites and proved wide-area distributed computation feasible (Foster et al., 1998). The Globus Project translated that insight into a reusable middleware toolkit that became the de facto standard through the 2000s (Foster & Kesselman, 1997). Europe responded with the EGEE projects and the European Grid Infrastructure (EGI); the United States built TeraGrid, XSEDE, and now ACCESS (Boerner et al., 2023). Today the Worldwide LHC Computing Grid (WLCG) links more than 170 computing centres in 42 countries under a single shared workload (ATLAS Collaboration, 2025). Grid computing is defined as the coordinated use of geographically dispersed, heterogeneous computational resources — owned by independent institutions — to solve problems beyond the capacity of any single system. Three foundational concepts structure every grid deployment: the Virtual Organisation (VO), which governs resource-sharing agreements; the meta-scheduler, which allocates work across VO boundaries; and the Grid Security Infrastructure (GSI), which enforces cross-institutional authentication and authorisation (Foster et al., 1998).

Problem Statement

Federating genuinely autonomous institutions creates challenges that neither cluster computing nor commercial cloud computing is designed to solve. Scheduling must span multiple administrative domains each enforcing independent priorities — a problem Hegde et al. (Hegde et al., 2024) prove is NP-complete in its general multi-objective form. Security must extend across domain boundaries without any single authority controlling all participants; the original X.509 proxy-delegation model has proven robust but operationally burdensome at scale, motivating an ongoing transition to OAuth 2.0 token-based authentication across WLCG and EGI (Dack et al., 2024). Data volumes

are growing faster than network bandwidth: the WLCG already manages hundreds of petabytes, and the Square Kilometre Array will generate 300 petabytes per telescope per year (Scaife, 2020). Environmental sustainability adds a further constraint: WLCG's 170 data centres consume substantial electrical power, and operating the HL-LHC programme within carbon-neutral limits will require simultaneous improvements across software, hardware, and energy procurement (Britton et al., 2024). These are not isolated problems but structural consequences of the federated architecture that makes grid computing institutionally viable.

Scope

This paper provides a comprehensive, analytically structured examination of Distributed Grid Computing across four sections. Section I analyses grid architectures: from the five-layer hourglass model (Foster et al., 2001) and the Open Grid Services Architecture through to contemporary EOSC-scale federated and microservices designs. Section II examines resource management and scheduling: multi-domain meta-scheduling, AI-enhanced job placement, data locality-aware scheduling using the Rucio/PanDA production stack, and scientific workflow management under FAIR reproducibility requirements. Section III covers the evolution of grid security and middleware: from X.509/GSI and VOMS authorisation to token-based federated identity, alongside analysis of the Globus, DIRAC, HTCondor, and UNICORE platforms. Section IV synthesises current challenges, presents case studies from high-energy physics, gravitational wave astronomy, genomics, climate science, and grid access in Kenya, and analyses grid–cloud–edge convergence. The paper argues that the federated governance model at grid computing's core — enabling independent institutions to share resources without surrendering autonomy — is the defining architectural insight that commercial cloud computing is structurally unable to replicate, and which makes grid infrastructure irreplaceable for large-scale scientific discovery.

Grid Computing Architectures

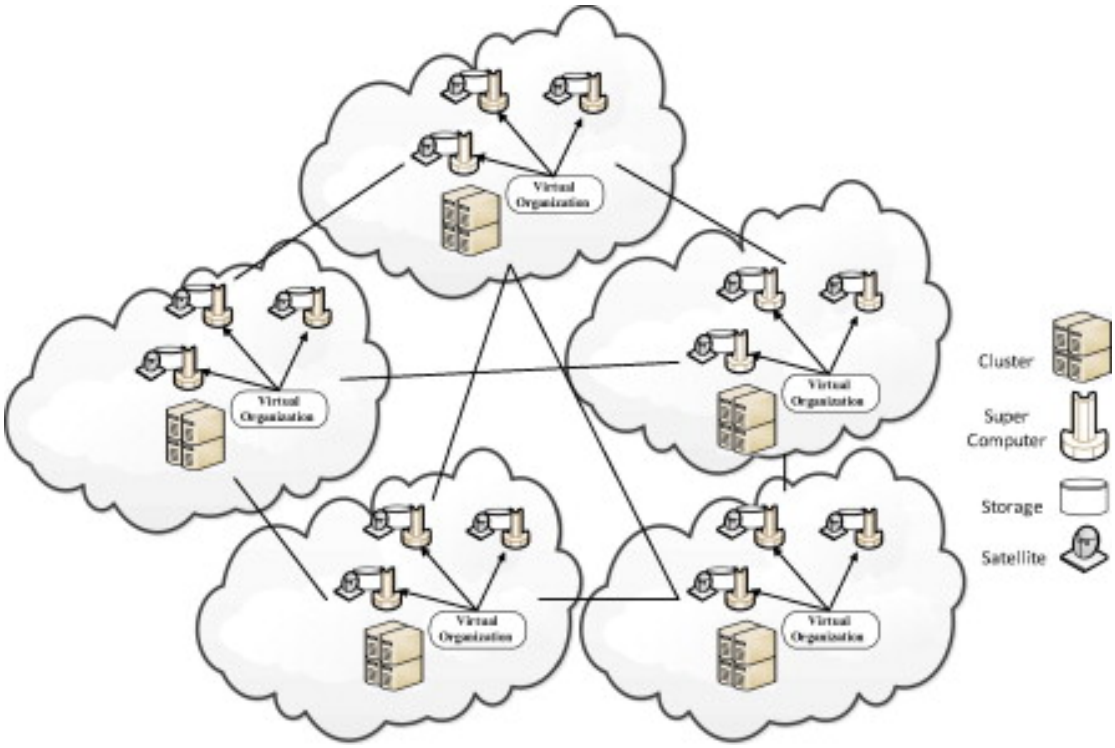


Figure 1: Grid Computing Architecture Overview

The Architgitectural Design Challenge

Designing an architecture for Distributed Grid Computing requires resolving a fundamental tension: the architecture must be flexible enough to accommodate the enormous heterogeneity of grid resources and use cases, yet standardised enough to enable interoperability across administrative boundaries. An architecture that is too rigid will fail to accommodate the diversity of real-world grid deployments; one that is too flexible will produce incompatible implementations that cannot interoperate. Foster and Kesselman (Foster & Kesselman, 1997) identified this tension in the earliest formulation of the Globus system, describing the need for a "vertically integrated treatment of application, middleware, and network" that could provide basic mechanisms such as authentication, resource discovery, and data access across heterogeneous systems — precisely the balance between standardisation and flexibility that continues to define grid architectural debates today.

The evolution of grid architectures over three decades reflects successive attempts to strike this balance. The five-layer hourglass model introduced through the Globus Toolkit established a minimal core of protocols at its "narrow waist", enabling both heterogeneous resources below and diverse applications above to interact through standardised interfaces (Foster, 2006; Foster & Kesselman, 1997). As web services technology matured, the Open Grid Services Architecture (OGSA) unified the grid resource model with WSDL-based web services, allowing grid resources to be exposed through standardised interfaces across heterogeneous environments (Foster et al., 2002; Snelling et al., 2009). Most recently, the European Open Science Cloud (EOSC) has emerged as the most ambitious expression of the federated architecture ideal: a "system of systems" approach that federates data, computing services, and research tools from institutions across Europe under a common interoperability framework, without any single entity controlling the underlying resources (European Commission, Directorate-General for Research and Innovation, 2022). Grigoryeva and Klimentov (Grigoryeva & Klimentov, 2024), in a comprehensive review published in Supercomputing Frontiers and Innovations, trace this thirty-year architectural trajectory and argue that each evolutionary step — from Globus to OGSA to EOSC — has preserved the core principle of minimal standardisation at the interface while progressively expanding the diversity of resources and use cases the architecture can accommodate.

The architectural design challenge is not merely a technical one: it is equally a governance challenge. Interoperability across administrative boundaries requires not only common protocols, but also institutional agreements about what those protocols mean and how they will be enforced. Snelling et al. (Snelling et al., 2009), in the IEEE Distributed Systems Online primer on OGSA, emphasise that achieving real-world interoperability demands standards-based components that allow architects to "meet their organisations' needs with confidence that they are interoperable with other standards-based grids, and that they can grow and adapt as those needs change." This insight — that architectural standardisation must be accompanied by institutional commitment to those standards — has shaped the governance structures of every major production

grid infrastructure from the WLCG to the EGI, and remains the central challenge for EOSC as it attempts to federate an unprecedentedly diverse ecosystem of European research services (European Commission, Directorate-General for Research and Innovation, 2022).

The Five-Layer Hourglass Architecture

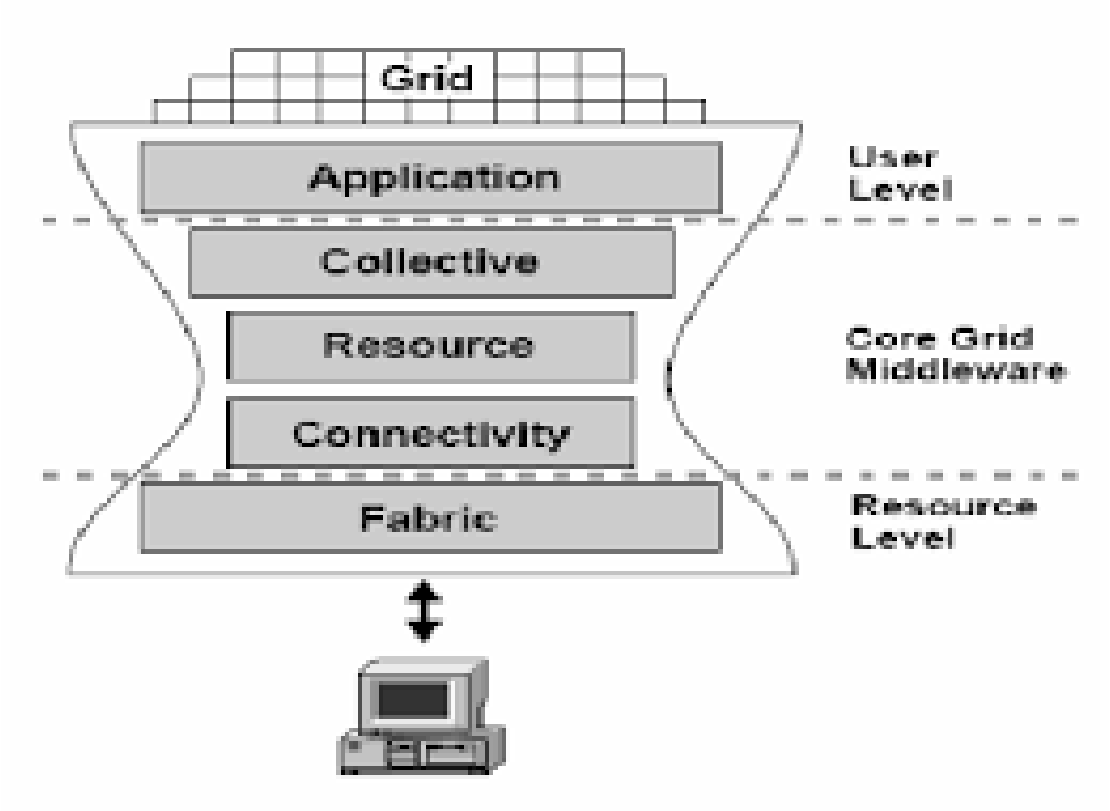


Figure 2: Original Hourglass Network Architecture

Section 1.1 established that the central challenge of grid architecture is resolving the tension between flexibility and standardisation. The five-layer hourglass model, formally articulated by Foster, Kesselman, and Tuecke (Foster et al., 2001)in their foundational paper "The Anatomy of the Grid," is the architecture that directly addresses this tension. The hourglass metaphor is deliberate and precise: a small, essential set of protocols at the narrow waist of the hourglass connects a diverse collection of resources below to a diverse collection of applications above. Foster et al. (Foster et al., 2001) describe this narrow neck as consisting of the Resource and Connectivity protocol layers, which are "designed so that they can be implemented on top of a diverse range of

resource types, defined at the fabric layer, and can in turn be used to construct a wide range of global services and application-specific behaviors at the collective layer." This minimalist core is precisely what allows the architecture to accommodate heterogeneity on both sides while enabling interoperability — the balance that Section 1.1 identified as the defining design challenge of the field. Grigoryeva and Klimentov (Grigoryeva & Klimentov, 2024) confirm that this hourglass principle has remained the defining structural feature of grid architectures across thirty years of evolution, surviving the transition from protocol-centric Globus to service-oriented OGSA to contemporary federated cloud deployments.

Fabric Layer

The Fabric layer comprises the physical and logical resources that grid computing aggregates: processors and computational clusters, mass storage systems, databases and scientific data archives, network infrastructure, and specialised instruments such as radio telescopes and genomic sequencers. As Foster et al. (Foster et al., 2001) describe it, "the basic Fabric elements are the same in each case: computers, storage systems, and networks." Each Fabric layer resource is managed by a local resource management system (LRMS) — also called a batch scheduler — that enforces site-specific access policies and executes submitted jobs. The LRMS is the point at which the grid meets institutional autonomy: it is the local administrator's tool for enforcing site policy, and the grid architecture explicitly respects rather than overrides it. Widely deployed LRMS platforms include SLURM (Simple Linux Utility for Resource Management), developed at Lawrence Livermore National Laboratory and now the dominant scheduler on Top500 supercomputers (Yoo et al., 2003); HTCondor, developed at the University of Wisconsin-Madison for high-throughput opportunistic workloads (Thain et al., 2005); PBS/TORQUE; and IBM's LSF. Critically, Fabric layer resources retain full autonomy under their local administrators: the hourglass architecture is designed so that participating sites do not surrender control of their resources to the grid — a design principle that distinguishes grid computing from cloud computing and is essential to securing

institutional participation (Foster et al., 2001).

Connectivity Layer

The Connectivity layer is the first of the two protocol layers that form the "narrow neck" of the hourglass — the standardised core that enables the entire architecture. It defines the core communication and authentication protocols that enable grid-specific network transactions across diverse administrative domains. (Foster et al., 2001) specify that this layer must satisfy three fundamental requirements: single sign-on, so that users authenticate once and access multiple resources without further interaction; mutual authentication, so that both user and resource verify each other's identity; and delegation, so that a service acting on a user's behalf can present the user's credentials to other services. These requirements are implemented through the Grid Security Infrastructure (GSI), which builds upon X.509 Public Key Infrastructure (PKI) standards. Foster, Kesselman, Tsudik, and Tuecke (Foster et al., 1998) designed GSI specifically to address the security requirements of multi-institutional grid environments, observing that conventional single-domain security systems cannot accommodate the combination of single sign-on, credential delegation, and site-controlled access policies required in a grid context. The protocols of the Connectivity layer are kept deliberately minimal — just enough to establish authenticated, encrypted communication channels — with all higher-level coordination delegated to the layers above, in accordance with the hourglass principle established in Section 1.1.

Resource Layer

The Resource layer, the second component of the hourglass's narrow neck, defines protocols for the negotiation, initiation, monitoring, control, accounting, and payment associated with sharing individual resources. It is important to note what this layer does not do: it deliberately restricts its scope to interactions with a single resource at a time, leaving the coordination of multiple resources to the Collective layer above. This deliberate constraint is a direct expression of the hourglass design principle identified in

Section 1.1 — by keeping the Resource layer minimal and single-resource-focused, the architecture ensures that any resource capable of implementing these protocols can participate in the grid, regardless of its type or local configuration. As Foster et al. (Foster et al., 2001) note in their description of the Anatomy architecture, the Grid Resource Allocation and Management (GRAM) enables grid jobs to be submitted to any participating site regardless of the site's underlying batch system, because the local LRMS is wrapped behind the standard GRAM interface — a concrete example of the hourglass principle in action.

Collective Layer

The Collective layer is where the distinctive complexity of grid computing resides — and where the architectural principle of the hourglass yields its greatest value. Because Collective layer components build on the standardised narrow neck of the hourglass, they can implement a wide variety of sharing behaviours without placing new requirements on the resources being shared (Foster et al., 2001). This is the key architectural insight: the standardised Resource and Connectivity layers act as a stable platform upon which an open-ended diversity of Collective services can be built, without requiring changes to the underlying fabric. Services at the Collective layer include directory and resource discovery services, which catalogue available resources using protocols such as GRIP and GRRP, as described by Czajkowski et al. (2001); co-allocation and advance reservation services, which reserve multiple resources simultaneously for jobs requiring coordinated access; workload management and meta-scheduling services, which distribute jobs optimally across available resources; data replication and management services; and monitoring and fault detection services. The Collective layer must accomplish global coordination without central control — each resource retains local autonomy under the Fabric layer's LRMS — making the design of scalable, robust Collective services one of the most actively researched problems in distributed systems (Grigoryeva & Klimentov, 2024).

Application Layer

The Application layer comprises the user-facing applications that exploit the grid through APIs and SDKs (Software Development Kits) exposed by the Collective and Resource layers. Foster et al. (Foster et al., 2001) illustrate this layer with two contrasting examples — a high-throughput ray-tracing rendering application and a multidisciplinary simulation — demonstrating that while the application-level requirements differ dramatically, both rely on the same underlying Fabric, Connectivity, Resource, and Collective layer services. Applications range from batch scientific simulation codes and parameter sweep studies to interactive data analysis portals and workflow management systems. The key architectural principle is transparency: applications interact with the grid through standard APIs and are insulated from the heterogeneity of the underlying fabric. As Grigoryeva and Klimentov (Grigoryeva & Klimentov, 2024) observe, this transparency principle proved essential to the long-term adoption of grid computing, because it allowed application developers to target the grid API rather than the diverse hardware beneath it. The five-layer hourglass architecture thus demonstrates in concrete architectural terms how the tension identified in Section 1.1 — between flexibility and standardisation — can be resolved: standardise minimally and precisely at the narrow neck, and leave both the resources below and the applications above free to evolve. This architectural logic is then extended and generalised in the Open Grid Services Architecture discussed in Section 1.3.

The Open Grid Services Architecture (OGSA)

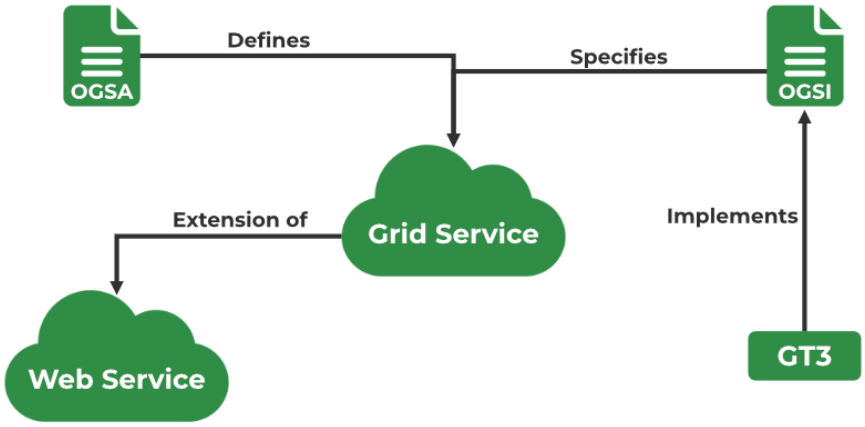


Figure 3: Open Grid Services Architecture

Section 1.2 demonstrated that the five-layer hourglass model deliberately keeps its protocol core minimal, enabling interoperability across heterogeneous resources without demanding conformance to a rigid framework. By the early 2000s, however, a new challenge had emerged: the growing dominance of web services technology in enterprise computing created pressure to align grid infrastructure with mainstream IT standards so that grid resources could participate in service-oriented enterprise architectures. The Open Grid Services Architecture (OGSA) was the direct response to this challenge. Formally introduced by Foster, Kesselman, Nick, and Tuecke (Foster et al., 2002) in their paper "Grid Services for Distributed Systems Integration," published in IEEE Computer, OGSA's central contribution was the unification of the grid resource model with the web services model: every grid resource — whether a computational cluster, a storage system, or a scientific instrument — is exposed as a grid service conforming to a standardised WSDL-defined interface. This alignment with mainstream IT standards enabled grid infrastructure to interoperate with enterprise service-oriented architectures, significantly broadening the potential ecosystem of grid adopters (Foster et al., 2002). Snelling et al. (Snelling et al., 2009), in their IEEE Distributed Systems Online primer on OGSA, confirm that this standards alignment was decisive in achieving

real-world interoperability: it allowed site architects to build standards-based grid components with confidence that they would be interoperable with other standards-based grids, and that the infrastructure could grow and adapt as organisational needs evolved.

A technically critical innovation that OGSA introduced was the concept of Stateful Resources — resources that maintain persistent state across multiple service invocations. This concept was essential because the conventional web services model, which OGSA was building upon, is inherently stateless: each service invocation is independent, and no state is preserved between calls. For grid computing, however, statelessness is fundamentally inadequate. Core grid entities such as running jobs, active data transfers, advance reservations, and workflow executions are inherently stateful: they are created at one point in time, evolve over hours or days, and must be consistently accessible and manageable throughout their lifetime. The technical framework for addressing this was the Web Services Resource Framework (WS-RF), proposed by Czajkowski, Ferguson, Foster, Frey, Graham, and Snelling (2004) as a set of web services specifications defining how to associate a Web service with one or more stateful resources and expose the resource's state through standard web service interfaces.

Foster, Czajkowski, Ferguson, Frey, Graham, Maguire, Snelling, and Tuecke (Foster et al., 2005), in their peer-reviewed paper "Modeling and Managing State in Distributed Systems: The Role of OGSI and WSRF," published in the Proceedings of the IEEE, provide the authoritative technical account of both the Open Grid Services Infrastructure (OGSI) — the first standardised implementation of OGSA's stateful service concept — and WS-RF, which refactored and evolved OGSI to better align with industry web services standards. They explain that both OGSI and WS-RF introduce "conventions on the use of the Web Services Definition Language schema that enable the modeling and management of state," with WS-RF providing the more mature and broadly adopted rendering. The importance of this cannot be overstated for grid computing: without statefulness, a grid infrastructure could not track a distributed workflow spanning multiple sites over days or weeks, could not maintain reservation agreements negotiated between scheduling systems, and could not guarantee the consistency of long-running

data transfers across wide-area networks — all of which are routine requirements in production scientific grids such as the WLCG (Grigoryeva & Klimentov, 2024).

Federated and Microservices Architectures

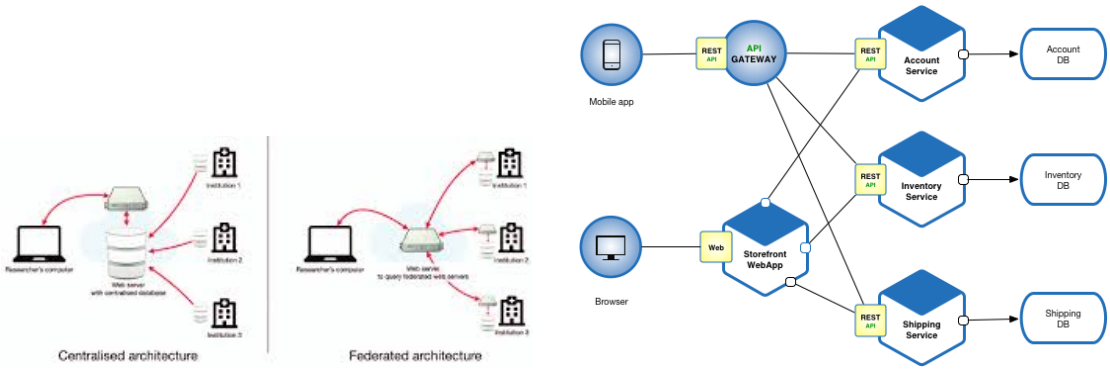


Figure 4: Federated and Microservices Architecture

Section 1.3 showed that OGSA's service-oriented approach, while technically sound, produced complex SOAP/WS-RF-based middleware that proved difficult to deploy at scale across heterogeneous institutional environments. Contemporary production grid infrastructures have consequently evolved in two complementary directions: toward federated governance models that formalise multi-stakeholder coordination, and toward microservices architectures that reduce operational complexity by decomposing monolithic middleware into independently deployable service components. The European Grid Infrastructure (EGI) exemplifies the federated model at its most mature. EGI's National Grid Initiatives (NGIs) are autonomous, officially recognised bodies that ensure the operation of grid infrastructures in their own country while representing the needs of their scientific communities within the broader federation. EGI currently federates 39 National Grid Initiatives from European countries and European Intergovernmental Research Organisations. The critical architectural principle is that each NGI retains full sovereignty over its own resources and policies while conforming to common interoperability standards — exactly the tension between autonomy and standardisation that Section 1.1 identified as the defining challenge of grid architecture. Calatrava et al. (2023), in their peer-reviewed survey of EOSC services published in Computer Science Review, document how this federated governance model has been progressively

extended beyond EGI toward the European Open Science Cloud, where federating institutions retain local control while participating in a common open science infrastructure governed through agreed interoperability frameworks.

The second direction of evolution — the adoption of microservices patterns from commercial cloud platforms — addresses a different problem: the operational complexity that accumulated in monolithic grid middleware stacks such as the original Globus Toolkit, gLite, and early UNICORE. Velepucha and Flores (2023), in their comprehensive survey published in IEEE Access, define microservices architecture as a style in which an application is decomposed into small, independently deployable services, each responsible for a distinct function and communicating through lightweight APIs. They identify the core architectural benefits as independent scalability, independent deployment, and technology heterogeneity — properties that are particularly attractive for grid middleware because different grid services (data transfer, job submission, authentication, workflow orchestration) have very different scaling and reliability requirements. The Globus platform, evolved from the monolithic Globus Toolkit through successive service-oriented redesigns, now exemplifies this approach in the scientific computing context: it is delivered as a set of independently deployable cloud-hosted services accessible through REST APIs, encompassing Globus Transfer, Globus Auth, Globus Compute, and Globus Flows (Chard, Pruyne, McKee, Bryan, Raumann, Ananthakrishnan, Foster, 2023). Chard et al. (2023), in their paper "Globus Automation Services" published in Future Generation Computer Systems, document how this microservices architecture enables research process automation across distributed infrastructure — precisely the kind of multi-site, long-running scientific workflow management that the Collective layer of Section 1.2's hourglass architecture was designed for, but now implemented through cloud-native REST APIs rather than SOAP-based grid services. This architectural evolution has dramatically lowered the learning curve for application developers by replacing complex, heavyweight WS-RF service interfaces with familiar REST/JSON APIs that integrate naturally with modern programming environments. Alongside the shift to microservices, production grid deployments have increasingly

adopted containerisation — packaging grid software and scientific applications in Apptainer (formerly Singularity) or Docker containers — to address the software dependency management challenges that historically complicated grid deployments across heterogeneous operating environments. Kurtzer, Sochat, and Bauer (2017), whose Singularity work has evolved into the Apptainer project now used across major HPC and grid sites, demonstrated that user-space containers can provide reproducible, portable execution environments without requiring root privileges — a critical constraint in multi-user grid environments where users cannot modify the host operating system.

The European Open Science Cloud and the GLUE Schema

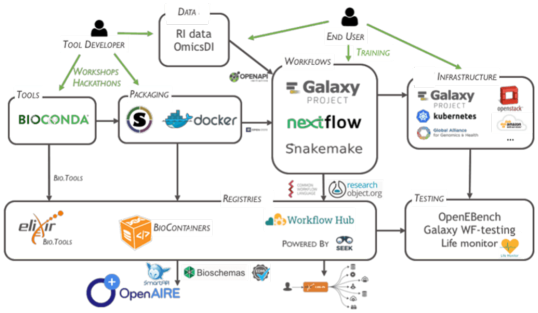


Figure 5: Europian Open Science Cloud

Section 1.4 described how federated governance models and microservices architectures have enabled the progressive integration of heterogeneous computing and data resources across institutional boundaries. The European Open Science Cloud (EOSC) represents the most ambitious current expression of this architectural trajectory. Formally initiated by the European Commission in 2016 and entering operational service through the EOSC EU Node launched in October 2024, EOSC aims to provide European researchers with an open, trusted, multi-disciplinary environment where they can publish, find, and reuse data, tools, and services for research and innovation (European Commission, Directorate-General for Research and Innovation, 2022).

EOSC is explicitly architected as a federation — described in official European Commission documentation as a “system of systems” — in which national research infrastructures, e-infrastructures such as EGI and GEANT, and thematic science clusters con-

tribute services and data repositories that are federated under a common governance framework and interoperability standard, without any single entity controlling the underlying resources. Calatrava et al. (Calatrava et al., 2023), in their peer-reviewed survey of EOSC services published in Computer Science Review, document how the EOSC-Synergy project — involving more than 20 institutions from eight European countries — integrated thematic scientific services across four domains (Earth Observation, Environment, Biomedicine, and Astrophysics) into the EOSC federation, identifying federated authentication, FAIR data management, and cloud-agnostic resource management as the three key architectural requirements for thematic service integration. The EOSC infrastructure consists of three interconnected technical layers: EOSC Core (providing cross-cutting services including single sign-on, resource catalogues, knowledge graphs, workflow management, and helpdesk); EOSC Exchange (offering domain-facing services including compute, containers, data transfer, and notebooks); and the EOSC Interoperability Framework (defining the protocols, metadata formats, and standards that participating services must adopt to achieve cross-domain and cross-infrastructure interoperability) (European Commission, Directorate-General for Research and Innovation, 2022).

Interoperability across heterogeneous grid middleware platforms requires that resources describe themselves using a common, middleware-independent vocabulary. This requirement — recognising that a grid site running UNICORE, an EGI site running the ARC middleware, and a Globus-connected HPC cluster must all be able to describe their capabilities in a mutually intelligible format — motivated the development of the GLUE (Grid Laboratory Uniform Environment) information schema through the Open Grid Forum (OGF). The authoritative specification is GLUE 2.0, documented in OGF Recommendation GFD.147, whose full authorship and formal citation is: Andreozzi, Burke, Ehm, Field, Galang, Konya, Litmaath, Millar, and Navarro (Andreozzi et al., 2009). GLUE 2.0 is designed as a conceptual information model — expressed in natural language enriched with UML Class Diagrams — that is deliberately independent of any concrete data model or serialisation format, with separate rendering specifications

for LDAP Schema (GFD.148), XML Schema, and JSON Schema (Memon et al., 2016) provided as companion documents.

The schema defines standardised entities for computational resources (ComputingService, ComputingShare, ComputingEndpoint), storage systems (StorageService, StorageEndpoint, StorageShare), software environments (ApplicationEnvironment, ApplicationHandle), network connectivity, and operational status attributes such as availability windows and capacity metrics. Burke, Field, and Horat (Burke et al., 2011), reporting the production migration to GLUE 2.0 across the LCG/EGEE/EGI grid infrastructure, document the complexity of adopting a new information model in a large, live production grid while maintaining backwards compatibility — a challenge that directly illustrates the governance dimension of architectural interoperability standards identified in Section 1.1. As of 2024, the GLUE schema continues to evolve through the OGF GLUE Working Group to accommodate cloud-native and container-based resources, reflecting the hybrid grid-cloud nature of contemporary research computing environments; the EOSC Interoperability Framework explicitly references GLUE 2.0 as a foundational standard for resource description within the federation (European Commission, Directorate-General for Research and Innovation, 2022).

Resource Management and Scheduling

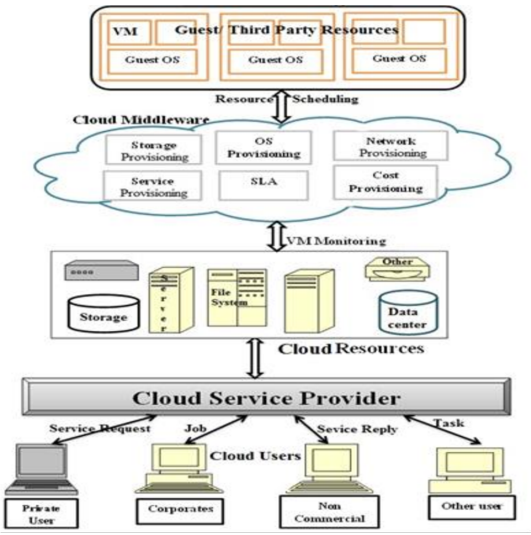


Figure 6: Resource Management and Scheduling in Cloud Environment

Section I established the architectural principles that enable Distributed Grid Computing: the five-layer hourglass model provides a minimal, stable protocol core across heterogeneous resources; OGSA extended this core with stateful, standards-based service interfaces; and federated architectures such as EGI and EOSC have formalised the multi-stakeholder governance frameworks necessary for large-scale resource sharing. Section II confronts the operational reality that these architectures create: given a federation of heterogeneous resources under diverse administrative authorities, how are computational tasks actually matched to appropriate resources, submitted, executed, and tracked? This is the resource management and scheduling problem in grid computing, and it is qualitatively more complex than its equivalents in cluster or cloud environments. In a cluster, a single scheduler has full visibility and authority over all resources. In a cloud, the provider controls the allocation layer. In a grid, the scheduler must operate across multiple autonomous domains simultaneously — each with its own batch system, local priorities, access policies, and failure modes — without being able to override any of them.

Hegde, Srinivas, Rajan, and colleagues (Hegde et al., 2024), in their open-access paper published in Scientific Reports, formalise this as an NP-complete multi-objective optimisation problem involving makespan, cost, and energy consumption simultaneously across heterogeneous grid nodes — demonstrating analytically why hand-crafted heuristics have historically dominated grid scheduling and why the emerging application of machine learning offers genuine advantages over rule-based approaches. Similarly, Yujian Wu and colleagues (Wu et al., 2025), in their comprehensive survey of task scheduling in geo-distributed computing published in IEEE Transactions on Parallel and Distributed Systems, identify the combination of heterogeneous network conditions, region-specific resource policies, and geographically varying computational capabilities as the defining structural challenges that set grid and geo-distributed scheduling apart from single-site or cloud scheduling — precisely the combination of properties that the federated architectures of Section I are designed to accommodate at the infrastructure layer.

The Multi-Domain Scheduling Problem

Resource management in grid computing differs qualitatively from resource management in cluster or cloud environments because it must operate across multiple administrative domains, each with its own local resource manager, access policies, and scheduling objectives. A grid meta-scheduler must solve a hierarchical optimisation problem: at the global level, it must allocate jobs to sites in a manner that achieves good overall performance while respecting each site's local constraints; at the local level, each site's batch scheduler must integrate grid-submitted jobs into its existing local workload. These two levels of scheduling can conflict, particularly when sites enforce local priority policies that preempt or delay grid jobs. Four structural properties make this problem qualitatively distinct from single-domain scheduling: autonomy (sites control their own resources and cannot be compelled to accept grid jobs); heterogeneity (sites differ in hardware, software, network connectivity, and local workload patterns); dynamism (resource availability changes continuously due to local workload fluctuations, maintenance, and failures); and policy conflict (globally optimal schedules routinely conflict with locally enforced priority and fairness policies). (Turilli et al., 2018), in their comprehensive analysis of distributed scientific computing scheduling published in ACM Computing Surveys, document precisely how these four properties combine to make grid-scale job management fundamentally different from single-site scheduling, showing that production systems such as OSG and WLCG process millions of jobs per day across heterogeneous cyberinfrastructures only because pilot-job architectures absorb the dynamism and policy conflicts at the middleware layer rather than exposing them directly to user-level schedulers. Hegde et al. (Hegde et al., 2024) further formalise the consequence: the combined multi-objective nature of the scheduling problem — optimising makespan, cost, and energy simultaneously across heterogeneous sites — is NP-complete in its general form, confirming analytically why no single scheduling heuristic can be universally optimal and why the field has progressively turned toward adaptive, AI-enhanced approaches discussed in Section 2.3. All four properties are direct consequences of the federated governance models described in Section 1.4 — the

same autonomy that makes federation attractive to participating institutions is precisely what makes scheduling across federations difficult.

Grid resource management is further complicated by the dynamic nature of grid resource availability. Sites join and leave the grid, maintenance windows take computational resources offline unexpectedly, and network conditions affecting data staging performance change continuously. Effective grid resource management therefore requires not only good scheduling algorithms but also robust mechanisms for real-time resource monitoring, failure detection, and adaptive job re-submission. Hegde et al. (Hegde et al., 2024) demonstrate that the computational intractability of the multi-objective grid scheduling problem — which they prove is NP-complete in its general form — means that exact optimisation is computationally infeasible for real-world grid scales, and that approximation algorithms and heuristics must be relied upon in practice. The consequence is that grid schedulers must balance solution quality against computational tractability, accepting near-optimal schedules that can be computed in real time over optimal schedules that cannot. (Wu et al., 2025) further note that in geo-distributed settings, network-level dynamics — varying bandwidth, latency, and data transfer costs across wide-area links — introduce an additional scheduling dimension absent from single-site systems, because the cost of moving data to a computation site can exceed the cost of running the computation itself. This is the data locality problem that Section 2.4 addresses in detail, and it illustrates how the scheduling challenges of Section II are structurally rooted in the architectural choices of Section I: federating geographically distributed resources under independent administrative control inevitably creates the data-compute co-location problem that distinguishes grid scheduling from all simpler variants of the resource management problem.

Resource Discovery and Information Services

The prerequisite for any scheduling decision is knowledge of what resources are currently available. Grid information services provide this knowledge by maintaining catalogues of resource availability, capability, and configuration. The Monitoring and

Discovery Service (MDS) in the Globus Toolkit provides a hierarchical LDAP-based directory of resource information: resources register their status with local aggregators that propagate summaries up a hierarchy to global information services, using the GLUE 2.0 information schema described in Section 1.5 to express their capabilities in a middleware-independent format. This hierarchical LDAP architecture was adequate for the grids of the early 2000s, but its polling-based design introduces a fundamental limitation: information staleness proportional to the polling interval. A scheduler working with information that is minutes old in a grid where jobs complete in seconds, or where sites fail without warning, will routinely make suboptimal or invalid placement decisions. This limitation became critical as grid infrastructure scaled from dozens of sites to hundreds.

The transition to event-driven, time-series monitoring architectures has been the production response to this limitation. Rather than polling resources at fixed intervals, event-driven systems propagate resource state changes in near-real time as they occur, using message brokers such as Apache Kafka and ActiveMQ to fan out state change events to all interested consumers simultaneously. The WLCG's MONIT infrastructure — the unified monitoring architecture for the CERN data centres and the WLCG grid — exemplifies this approach at production scale. Aimar, Aguado Corman, Andrade, Delgado Fernandez, Garrido Bear, Karavakis, Marek Kulikowski, and Magnoni (Aimar et al., 2019), in their open-access paper published in EPJ Web of Conferences, document how MONIT replaced legacy in-house and commercial monitoring solutions with a fully open-source stack built on Collectd, Kafka, Spark, Elasticsearch, InfluxDB, and Grafana — precisely the DevOps monitoring tools now standard in commercial cloud operations — to cover the complete monitoring data workflow from metric collection and validation through to dashboards, reports, and alarms. The significance of this architectural choice for grid scheduling is direct: by replacing polling with event streaming, MONIT reduces the window during which a scheduler might dispatch jobs to failed or overloaded resources from minutes (under polling) to seconds (under event streaming), translating directly into fewer failed job submissions and less wasted com-

pute time across the 170-site WLCG federation. The EGI independently deployed its ARGO monitoring framework on similar principles, using Prometheus and message brokers to provide availability and reliability metrics across its federated infrastructure. Andreeva, Aimar, Kerekes, and Saiz (Andreeva et al., 2010), in the Journal of Grid Computing, provide the foundational production evidence for dashboard-based monitoring of LHC Virtual Organisation computing activities — the operational monitoring model that MONIT evolved from — confirming that centralised, unified monitoring is an operational necessity rather than an optional enhancement for grids of WLCG's scale.

Scheduling Algorithms and AI-Enhanced Optimisation

Grid scheduling algorithms must balance multiple competing objectives simultaneously: minimising job turnaround time, maximising resource utilisation, respecting priority agreements between Virtual Organisations (VOs) and resource providers, and satisfying data locality constraints. The problem is structurally different from single-site scheduling because, as established in Section 2.1, the scheduler cannot force any participating site to accept jobs or alter its local policies. Classic grid scheduling approaches include list scheduling (ranking jobs by priority and assigning each to the best available resource), backfill scheduling (filling schedule gaps with smaller jobs whose execution will complete before higher-priority jobs require those resources), and advance reservation (allowing high-priority jobs to reserve resources at a specified future time). Turilli, Santcroos, and Jha (Turilli et al., 2018), in their ACM Computing Surveys analysis of pilot-job systems — the dominant paradigm for production grid job management — explain that these classic approaches must be implemented at the meta-scheduler level through the pilot-job abstraction precisely because the grid meta-scheduler cannot directly control local batch systems: pilot jobs act as placeholders that claim local resources, with the actual work units injected dynamically after the pilot has acquired a slot. This indirection is not an inefficiency — it is the architectural solution to the policy conflict and dynamism properties identified in Section 2.1.

A significant and well-documented development since 2022 has been the application of machine learning techniques to scheduling optimisation. Adimora, Gundla, and Sun (Adimora et al., 2025), in their comprehensive survey of machine learning approaches for HPC scheduling published in Cluster Computing, systematically review supervised learning, unsupervised learning, reinforcement learning, and neural network approaches across job allocation, workload prediction, load balancing, energy efficiency, and fault tolerance — reporting reductions in job completion time of up to 30% and improvements in resource utilisation of up to 25% compared to traditional heuristics. For the specific problem of runtime prediction — which directly enables more accurate backfill scheduling and advance reservation — Zrigui, de Camargo, Legrand, and Trystram (Zrigui et al., 2022), in a paper published in the Journal of Parallel and Distributed Computing, demonstrate that a simple online classifier trained on recent workload history and used to distinguish short from long jobs outperforms both static policies and previous regression-based runtime prediction approaches, with the key advantage that the online approach requires only weeks of historical data rather than the months needed by regression models. For reinforcement learning applied directly to grid-scale scheduling, Casagrande, Koslovski, Miers, Pillon, and Gonzalez (Casagrande et al., 2022), in the International Journal of Grid and Utility Computing, demonstrate that a deep RL agent can learn energy-aware shutdown scheduling policies that achieve up to 7% energy savings and 27% reduction in average job waiting time compared to static rule-based policies — confirming that RL can acquire grid-specific scheduling knowledge that is difficult to encode as hand-crafted rules. These AI-enhanced scheduling approaches are progressively being integrated into production grid meta-schedulers including HTCondor and SLURM, representing a significant evolution in grid resource management capability beyond the classical heuristics that have dominated production deployments for two decades.

Economic models for grid scheduling have also gained renewed attention as production grids have increasingly interfaced with commercial cloud resources. In a computational economy, jobs are submitted with associated budgets and resource providers set prices

based on supply and demand; auction mechanisms, commodity market models, and contract-based models have been studied in both theoretical and experimental grid contexts. The practical realisation of these economic scheduling ideas has come primarily through hybrid grid-cloud architectures, where commercial cloud providers offer on-demand resources at published prices alongside the fixed-capacity grid, allowing meta-schedulers to make economically rational placement decisions based on job priority, deadline, and cost budget. Chard, Pruyne, McKee, Bryan, Raumann, Ananthakrishnan, Chard, and Foster (Chard et al., 2023), in their paper on Globus Automation Services published in Future Generation Computer Systems, document how the Globus platform now implements multi-site workflow orchestration that can direct work to grid sites, institutional HPC resources, or commercial cloud providers within the same workflow execution — the operational instantiation of the economic scheduling model in production research computing. This hybrid scheduling capability is analysed in depth in Section IV in the context of ACCESS and the NSF's on-ramp programme for commercial cloud integration.

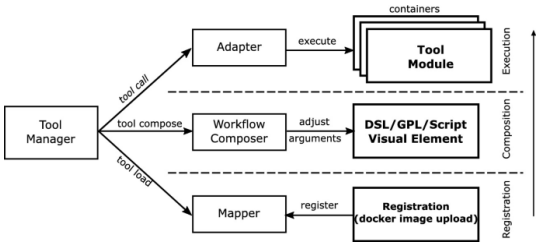
Data Locality and the Data Grid Scheduling Problem

One of the most significant performance challenges in data-intensive grid computing is the data locality problem: computation is fastest when the data it requires is co-located on the same system. In a grid environment, data may be stored at one site and compute resources at another, requiring data transfer over the wide-area network before computation can begin. For large datasets — and in contemporary particle physics, genomics, and climate science, datasets routinely run to hundreds of petabytes — this transfer time can dominate total job time, making the choice of which compute resource to use critically dependent on where the required data is already located. As Wu et al. (Wu et al., 2025) note in their survey of task scheduling in geo-distributed computing, network-level dynamics in wide-area environments — varying bandwidth, latency, congestion, and data transfer costs across intercontinental links — introduce a scheduling dimension absent from single-site systems, because the cost of moving data to a computation site can exceed the cost of running the computation itself. This insight directly moti-

vates the data locality-aware scheduling approaches that characterise production data grids.

Scheduling systems for data-intensive grids must therefore consider data locality as a first-class scheduling objective alongside resource availability and queue length. The architectural solution is the combination of a distributed data management system that tracks the locations of all data replicas and implements replication policies, with a workload management system that uses replica location information as a primary input to its site selection algorithm. Barisits, Beermann, Berghaus, and colleagues (Barisits et al., 2019), in the peer-reviewed paper "Rucio: Scientific Data Management" published in Computing and Software for Big Science, document how Rucio implements this through its concept of Rucio Storage Elements (RSEs) — logical abstractions over heterogeneous storage systems — and replication rules that specify how many replicas of a dataset must exist, at which sites, and for how long. Crucially, Rucio maintains a functional distance metric between RSEs that encodes effective transfer throughput rather than geographic distance: two storage systems with high-bandwidth connectivity are treated as "closer" than two geographically adjacent systems with network constraints, directly expressing the scheduling-relevant cost of data movement in the system's data model. Maeno, Alekseev, Barreiro Megino, De, Guan, Karavakis, Klimentov, Korchuganova, Lin, Nilsson, Wenaus, Yang, and Zhao (Maeno et al., 2024), in the authoritative PanDA system paper published in Computing and Software for Big Science, describe how the PanDA workload management system integrates Rucio replica information directly into its brokerage algorithm — selecting compute sites for each task based on a combination of site capacity, queue backlogs, and data availability — handling workloads at the exabyte scale across approximately 200 sites worldwide. The PanDA-Rucio integration represents the most mature production implementation of data locality-aware grid scheduling in existence, and its architecture has been extended beyond the ATLAS experiment to the CMS, ALICE, and LHCb experiments, as well as to the Vera C. Rubin Observatory's Legacy Survey of Space and Time, confirming its generality as a data locality scheduling framework for large-scale distributed science.

Workflow Management and Scientific Reproducibility

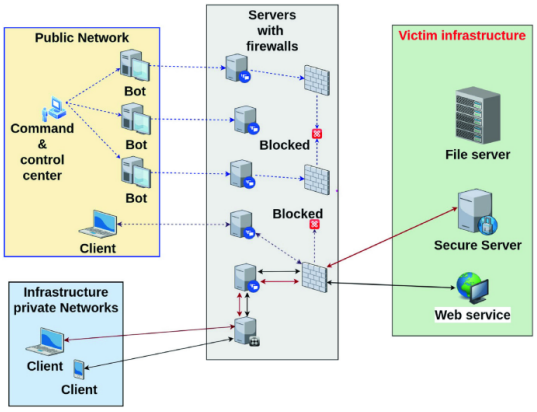


Scientific grid applications increasingly take the form of complex, multi-step workflows in which the output of one computational task serves as the input to the next. Managing such workflows in a grid environment requires systems capable of expressing task dependencies, coordinating data staging between tasks, handling partial failures, and tracking data provenance for scientific reproducibility. Deelman, Vahi, Rynge, Mayani, Ferreira da Silva, Papadimitriou, and Livny (Deelman et al., 2019), in the definitive evolution paper for the Pegasus Workflow Management System published in Computing in Science Engineering, document how Pegasus has since 2001 developed into a robust system that automates the execution of complex applications on heterogeneous, distributed high-throughput and high-performance computing environments.

A core principle of the Pegasus design — the separation of workflow description from workflow execution — is what enables scientific workflows to be mapped adaptively onto diverse target environments including campus clusters, national cyberinfrastructures, and commercial clouds. Pegasus handles data staging automatically: it inserts data transfer and registration jobs into the workflow graph as needed, selecting storage and compute resources based on data locality and site availability, and integrating with distributed data management systems such as Rucio to resolve replica locations before scheduling each task. Ferreira da Silva, Badia, Bala, Bard, and colleagues (Ferreira da Silva et al., 2023), in the community roadmap report emerging from the 2022 Workflows Community Summit, identify the continued evolution of workflow systems toward heterogeneous computing — spanning edge nodes, HPC centres, cloud, and quantum backends — as the defining challenge for the next generation of scientific workflow management.

The importance of workflow provenance — the ability to trace exactly what computation was performed, with what software and parameters, on what input data — has grown significantly as reproducibility has become a central concern of scientific publishing. The W3C PROV standard and its extension to the scientific computing context through the ProvONE model provide formal frameworks for representing computational provenance. Goble, Cohen-Boulakia, Soiland-Reyes, Garijo, Gil, Crusoe, Peters, and Schober (Goble et al., 2020), writing in the journal Data Intelligence, establish the case that workflow systems are not merely tools for executing computation but are themselves digital objects that must satisfy FAIR principles — they must be findable via registries with persistent identifiers, accessible through standardised protocols, interoperable across workflow engines, and reusable through rich metadata describing their inputs, parameters, and computational environment. Wilkinson, Dumontier, Aalbersberg, and colleagues (M. D. Wilkinson et al., 2016), in the foundational FAIR principles paper published in Scientific Data, first articulated the framework of Findable, Accessible, Interoperable, and Reusable requirements that now underpins data management mandates from the European Commission, the US National Science Foundation, the US National Institutes of Health, and major national funding agencies worldwide. Extending this to software and workflows specifically, Wilkinson, Aloqalaa, Belhajjame, Crusoe, and colleagues (S. Wilkinson et al., 2025), in a follow-on paper published in Scientific Data, confirm that the FAIR principles apply directly to computational workflows — requiring that workflows be deposited in searchable registries, assigned globally unique identifiers, described with sufficiently rich metadata to enable reuse, and executed in reproducible software environments. Together, these requirements establish that integration of provenance capture, workflow registration, and FAIR-compliant metadata publication into workflow management systems is now considered a minimum standard for scientific grid infrastructure.

Security Infrastructure and Middleware

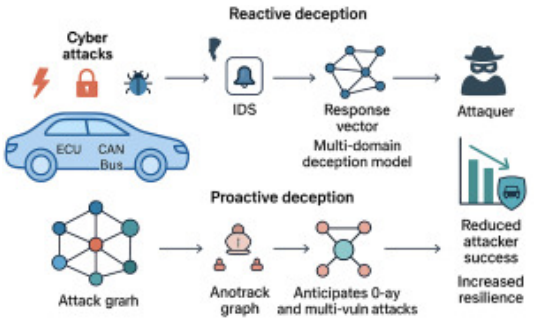


The architectural foundations established in Section I and the resource management mechanisms examined in Section II are both entirely dependent on a third dimension that makes production-scale grid computing possible: security. A grid environment that federates hundreds of administrative domains, processing millions of jobs across petabytes of data belonging to thousands of researchers worldwide, operates in an adversarial environment in which authentication failures, authorisation bypasses, or compromised credentials can cascade across institutional boundaries in ways impossible in single-domain systems. Section III analyses the evolution of grid security infrastructure — from the original Grid Security Infrastructure (GSI) designed by Foster, Kesselman, Tsudik, and Tuecke (Foster et al., 1998) and discussed in Section 1.2 through to the token-based federated identity systems replacing X.509 proxy certificates in contemporary production grids — and examines the major middleware platforms that translate the architectural principles of Sections I and II into deployable software. The GSI design choices made in 1998 directly shaped the architectures of every production grid that followed: WLCG, EGI, OSG, and EOSC all carry the imprint of the GSI's PKI-centred trust model.

The contemporary transition toward OAuth 2.0, OpenID Connect, and capability tokens represents not a rejection of the GSI's principles but an evolution of its implementation, driven by the usability barriers and scalability limits of X.509 certificate management at exascale. Withers, Bockelman, Weitzel, Brown, Gaynor, Basney, Tannenbaum, and

Miller (Withers et al., 2018), in the SciTokens paper presented at PEARC ’18, document precisely this transition: whereas X.509 proxy certificates authenticate identity and then leave authorisation to site-level policies, capability-based OAuth tokens embed the specific authorisations required by each workflow directly into the token itself, eliminating the credential management overhead that historically plagued large-scale grid workflows. Understanding this evolution is essential for understanding why contemporary grid middleware — Globus, DIRAC, HTCondor, and UNICORE — looks the way it does, and why Section III’s analysis of security infrastructure and middleware must be read as a single unified account rather than two separate topics.

The Multi-Domain Security Problem



Graphical abstract — Reactive deception (IDS → response vector) and proactive (strategic placement of honeypots on the attack graph)

Security in Distributed Grid Computing is fundamentally more challenging than in single-domain distributed systems because it must span multiple administrative security domains, each with its own user directory, access control framework, and trust model. The challenge is not merely technical but structural: no single authority controls all the participating institutions, yet the security of any one site depends on the integrity of the credential infrastructure shared across all of them. A particle physicist at the University of Nairobi submitting a job to the WLCG must be authenticated by a certificate authority trusted by CERN, authorised by the Virtual Organisation’s policy, and then authenticated and authorised again by the local site’s security framework — all without manual intervention at each step and without revealing the user’s private credentials to any intermediate service.

Foster, Kesselman, Tsudik, and Tuecke (Foster et al., 1998), in the paper "A Security Architecture for Computational Grids" presented at ACM CCS and discussed in the Connectivity layer analysis of Section 1.2, designed the Grid Security Infrastructure (GSI) to satisfy exactly this requirement. They identified six non-negotiable security properties for a multi-domain grid environment: single sign-on (users authenticate once, not at each site); mutual authentication (both user and service verify each other's identity, preventing impersonation of either party); message integrity and confidentiality (data in transit is protected against eavesdropping and tampering); delegation (a service can act on a user's behalf without requiring the user's interactive presence); fine-grained authorisation (access decisions are based on rich policy rather than binary allow/deny); and accurate accounting (resource usage is attributable to specific identities for auditability and billing).

These six requirements, formulated for the grids of the late 1990s, remain equally applicable to the exascale federated infrastructures of 2024 — what has changed is the technology used to implement them. The X.509 PKI approach that the GSI adopted to satisfy these requirements was technically sound but carried a structural usability cost: obtaining and managing grid certificates requires interaction with a Grid Certificate Authority's registration process, certificate lifetimes must be actively managed, and proxy certificates must be renewed for long-running jobs. At the scale of tens of thousands of users across hundreds of institutions, this certificate management overhead constitutes a significant barrier to participation. Dack, Agostini, Basney, Cornwall, De Stefano, Dykstra, Giacomini, Litmaath, Miccoli, Sallé, Short, and Vianello (Dack et al., 2024), representing the WLCG Authorization Working Group in their CHEP 2023 paper published in EPJ Web of Conferences, document precisely this motivation: the WLCG Authorization Working Group was formed in July 2017 explicitly to address the limitations of the X.509-GSI model and to define a token-based replacement using modern OAuth 2.0 and OpenID Connect standards, with the objective of satisfying the same six security requirements as the original GSI while eliminating the certificate management overhead that had accumulated over two decades of production operation. The transi-

tion they describe — from identity-centric X.509 proxy delegation to capability-centric OAuth bearer tokens — is the most significant evolution in grid security infrastructure since the original GSI design, and its architecture is analysed in detail in the next sections 3.2 and 3.5.

X.509 Certificates and the International Grid Trust Federation

The GSI design builds upon the X.509 Public Key Infrastructure (PKI) standard, formally specified in the Internet Engineering Task Force RFC series — most authoritatively in RFC 5280 (Cooper et al., 2008), which defines the Internet X.509 certificate and Certificate Revocation List profile. Under this framework, every grid user and every grid service endpoint is issued an X.509 digital certificate by a trusted Grid Certificate Authority (Grid CA). The certificate cryptographically binds the holder's public key to their identity — expressed as a Distinguished Name (DN) — through a digital signature by the issuing CA, so that any relying party possessing the CA's public key can independently verify the binding without contacting the CA at authentication time. When a grid user authenticates to a resource, they prove possession of the private key corresponding to their certificate through a TLS or GSI-specific challenge-response exchange; the resource then validates the certificate chain and checks the Certificate Revocation List (CRL) or Online Certificate Status Protocol (OCSP) service to confirm the certificate has not been revoked.

Cornwall, Jensen, Kelsey, Frohner, Kouril, Bonnassieux, Nicoud, Lörentey, Hahkala, Silander, Cecchini, Ciaschini, dell'Agnello, Spataro, O'Callaghan, Mulmo, Volpato, Groep, Steenbakkers, and McNab (Cornwall et al., 2004), in their peer-reviewed paper on authentication and authorisation mechanisms for multi-domain grid environments published in the Journal of Grid Computing, document precisely how this chain-of-trust model was deployed across the European DataGrid and related infrastructures. They show that the critical challenge in multi-domain grid authentication is not merely the cryptographic mechanism — which standard X.509 provides — but the establishment and maintenance of a consistent, globally agreed list of trusted CAs across institu-

tions that have no pre-existing bilateral trust relationships, making the trust federation infrastructure as important as the cryptographic infrastructure itself. Alfieri, Cecchini, Ciaschini, dell'Agnello, Frohner, Lörentey, and Spataro (Alfieri et al., 2005), in the definitive peer-reviewed account of the VOMS authorisation service published in Future Generation Computer Systems, trace the evolution from the earliest grid authorisation model — the gridmap file, a static per-site mapping of certificate Distinguished Names to local Unix accounts — to VOMS, which encodes VO membership and role attributes directly into proxy certificate extensions. This architectural progression is essential context for section 3.2: the X.509 chain-of-trust establishes that a certificate is authentic; VOMS then uses that authenticated identity to determine what the certified user is permitted to do within a specific VO, separating authentication (who are you?) from authorisation (what may you do?) — a distinction that Section 3.5 will develop in full.

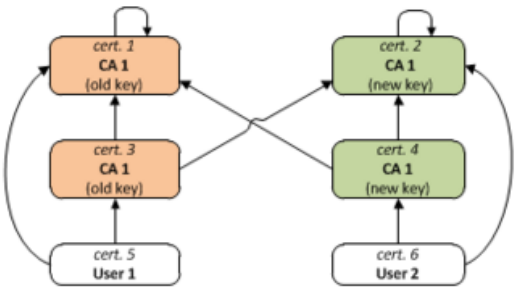
Proxy Certificates and Delegation

A central mechanism of GSI is proxy certificate delegation — the ability of a grid user to create a short-lived, derivative certificate (a proxy certificate) that authorises grid services to act on the user's behalf without requiring interactive presence for each operation. The technical specification for proxy certificates is RFC 3820, authored by Tuecke, Welch, Engert, Pearlman, and Thompson (Tuecke et al., 2004) as an IETF Internet Standards Track document. RFC 3820 defines a proxy certificate as one derived from, and signed by, a normal X.509 end-entity certificate or by another proxy certificate, for the purpose of providing restricted proxying and delegation within a PKI-based authentication system. Crucially, the RFC introduces the concept of restricted proxies — proxy certificates carrying policy extensions that limit the rights they convey to a strict subset of those held by the originating end-entity certificate — so that a workflow management system delegated a restricted proxy cannot exercise the user's full authorisations, limiting the blast radius of a credential compromise to only the specific operations the workflow requires. When a user submits a workflow to a grid, they generate a proxy certificate — typically with a lifetime of 12 to 48 hours — and delegate it

to the workflow management system.

The workflow system presents this proxy in GSI-secured TLS handshakes to authenticate job submissions, data staging operations, and service interactions across all participating grid sites, without the user being present at each step. Basney, Humphrey, and Welch (Basney et al., 2005), in the peer-reviewed paper "The MyProxy Online Credential Repository" published in Software: Practice and Experience, document how MyProxy addresses the operational challenge that proxy delegation creates for long-running grid jobs: since a proxy certificate has a fixed validity period of 12–48 hours, a workflow spanning multiple days would fail when its initial proxy expires unless a renewal mechanism exists. MyProxy solves this by providing a secure server-side repository in which the user deposits a longer-lived credential at job submission time, protected by a separate passphrase; the workflow agent retrieves freshly-delegated short-lived proxy credentials from MyProxy as needed throughout the job's lifetime. Critically, the private key stored in the MyProxy repository is never transmitted over the network — the client receives a newly-delegated proxy, not the private key itself — preserving the security properties of the GSI credential model while enabling the multi-day, multi-site grid workflows that particle physics, genomics, and climate science routinely require. The dependency of this entire proxy delegation infrastructure on the X.509 certificate lifecycle is precisely what motivates the token-based transition that we are going to discuss in Section 3.4: as Dack et al. (Dack et al., 2024) document, the WLCG's move toward OAuth 2.0 bearer tokens directly addresses the operational complexity and scalability limits of long-running proxy credential management that even the sophisticated MyProxy architecture could only partially mitigate.

Federated Identity: The Transition Beyond X.509



A significant security evolution in contemporary grid infrastructure is the transition from X.509 grid certificates toward federated identity and token-based authentication. X.509 certificates have historically been a significant usability barrier: obtaining a grid certificate requires interaction with a Grid CA's registration process, managing certificate lifetimes adds operational complexity, and the manual registration authority model scales poorly to the tens of thousands of researchers who routinely need grid access. The AARC (Authentication and Authorisation for Research and Collaboration) Blueprint Architecture, developed through successive European Commission-funded projects beginning in 2015, defines a framework for integrating institutional identity providers (using SAML or OpenID Connect protocols) with research authorisation infrastructure.

Biancini, Florio, Haase, Hardt, Jensen, Kanellopoulos, Liampotis, Licehammer, Memon, van Dijk, Paetow, Prochazka, Sallé, Solagna, Stevanovic, and Vaghetti (Biancini et al., 2016), in the first formal publication of the Blueprint Architecture (arXiv:1611.07832), establish the central design principle: rather than requiring every user to obtain credentials from every resource provider, the AARC model interposes a Community AAI layer that translates the user's existing institutional identity into the authorisation assertions each resource provider requires. The 2019 edition of the Blueprint Architecture, formally published by the AARC consortium on Zenodo (AARC – Authentication and Authorisation for Research and Collaboration, 2019), codifies this into a five-layer model and introduces the Community AAI component now deployed at EGI, EOSC, and WLCG as the standard federated access framework.

Under the AARC model, researchers authenticate using their institutional credentials through their national identity federation, and an attribute authority (such as VOMS or the AARC Group Manager) asserts their VO membership and role attributes to grid services as signed tokens. The transition to token-based authentication using JWT (JSON Web Tokens) and the WLCG Token-Based Authentication standard has been underway since 2022 across major grid infrastructures including the WLCG and EGI, and represents the most significant evolution in grid security since the original GSI design — as documented by Dack et al. (Dack et al., 2024), who describe the WLCG Authorization Working Group's seven-year programme to replace X.509 proxy delegation with capability-based tokens across all LHC experiments and computing sites.

The SciTokens project, described by Withers, Bockelman, Weitzel, Brown, Gaynor, Basney, Tannenbaum, and Miller (Withers et al., 2018) in their PEARC '18 paper published in the ACM Digital Library, established the capability-token model that now underpins this transition: by encoding authorisations directly into OAuth 2.0 JSON Web Tokens rather than relying on identity impersonation through X.509 proxy certificates, SciTokens allows grid workflows to convey precise, workflow-specific permissions — for example, read access to a specific dataset directory — to remote services while eliminating the risks of broad credential delegation. The normative specification for how SciTokens-style capability tokens are structured within the WLCG context is the WLCG Common JWT Profiles document (Altunay et al., 2019), published on Zenodo, which defines the token claims, issuer requirements, and validation procedures that every WLCG site must implement to accept bearer tokens from any authorised VO.

The CILogon federated identity service, documented by Basney, Fleury, and Gaynor (Basney et al., 2014) in Concurrency and Computation: Practice and Experience, integrates grid PKI with institutional identity providers using SAML and OpenID Connect federation standards; it serves as the token issuer for numerous US open-science communities and bridges the legacy grid security world with the contemporary federated identity framework, enabling researchers at institutions connected to InCommon or eduGAIN to obtain grid-compatible tokens using their existing institutional login

without any separate grid certificate registration.

Authorisation: VOMS, XACML, and Attribute-Based Access Control

Authentication establishes identity; authorisation determines what an authenticated identity is permitted to do — and in a multi-domain grid environment, these are deeply distinct problems. Grid authorisation has evolved through three generations, each adding expressiveness in response to the limitations of its predecessor. The first generation, the gridmap file, was a static, per-site text file maintained by local administrators that mapped each user's X.509 Distinguished Name to a local Unix account; it required manual updates at every site for every new user and was entirely incapable of expressing VO-level roles or fine-grained permissions. The second generation, VOMS (Virtual Organization Membership Service), addressed this by embedding VO membership and role attributes directly into signed extensions of proxy certificates, so that a single VOMS attribute certificate could carry the user's group memberships and roles to every site in the federation without per-site manual configuration. Alfieri, Cecchini, Ciaschini, dell'Agnello, Frohner, Lörentey, and Spataro (Alfieri et al., 2005), in the definitive peer-reviewed account of VOMS published in Future Generation Computer Systems — already cited in Section 3.2 — show that VOMS fundamentally changed the operational model of grid authorisation: rather than resource sites managing per-user access lists, the VO manages a central membership database and issues signed attribute assertions that resource sites honour through VOMS-aware access control frameworks.

The third generation, Attribute-Based Access Control (ABAC) using the XACML (eXtensible Access Control Markup Language) policy language, moves beyond VO membership to express rich combinations of user attributes (VO membership, role, institution, assurance level), resource attributes (resource type, data sensitivity, classification), and environmental conditions (time of day, network location, current load). Garzoglio, Alderman, Altunay, Ananthakrishnan, Bester, Chadwick, Ciaschini, Demchenko, Ferraro, Forti, and colleagues (Garzoglio et al., 2009), in their peer-reviewed paper on the definition and implementation of a SAML-XACML profile for authorisation interop-

erability across grid middleware in OSG and EGEE, published in the Journal of Grid Computing, document the first production deployment of XACML-based grid authorisation: by defining a common SAML-XACML attribute and obligation profile that both OSG and EGEE middleware could implement, this work enabled, for the first time, policy-based authorisation decisions that were consistent across two otherwise architecturally distinct grid infrastructures — demonstrating that ABAC is not merely a theoretical improvement over VOMS but a practically deployable, production-tested authorisation model for multi-domain scientific grids.

The INDIGO-IAM (Identity and Access Management) service represents the convergence of all three authorisation generations — identity federation, VO attribute management, and token-based access control — into a single production-deployed service. Ceccanti, Vianello, Caberletti, Costantini, Guarise, Italiano, Maraschini, Misurelli, Nardelli, Pace, and Ronchieri (Ceccanti et al., 2017), in the peer-reviewed paper "The INDIGO-DataCloud Authentication and Authorization Infrastructure" published in the Journal of Physics: Conference Series, describe how INDIGO-IAM was architected to replace the fragmented AAI landscape of pre-EOSC European grid infrastructure.

The system combines OIDC-based authentication with fine-grained group and role management, SCIM provisioning for account lifecycle management across services, and token exchange capabilities that allow a user's institutional OpenID Connect identity to be translated into the VO membership assertions and authorisation tokens that downstream grid services require. Critically, INDIGO-IAM implements account linking — the ability to associate multiple external identity credentials (OpenID Connect accounts from different providers, legacy X.509 certificates, and SSH keys) with a single persistent IAM identity — enabling the incremental migration from X.509-based grid access to token-based federated access without requiring users to abandon their existing credentials or workflows during the transition. The INDIGO-IAM software is maintained as an open-source project and was formally released on Zenodo (Agostini et al., 2024). As of 2024, INDIGO-IAM is deployed as the reference IAM service for

the EOSC and EGI federations, directly implementing the AARC Blueprint Architecture described in Section 3.4 — making it the production realisation of the complete three-generation authorisation evolution from gridmap files to attribute-based federated identity that this section has traced.

Major Middleware Platforms

The Globus Platform

The Globus platform, evolved from the Globus Toolkit through a decade of architectural redesign, is the most widely deployed grid and research computing middleware in the world as of 2025. Unlike its predecessor, which was distributed as open-source software that each site installed and operated independently, the contemporary Globus platform is delivered primarily as a cloud-hosted service with lightweight software agents (Globus Connect Personal and Globus Connect Server) deployed at each participating endpoint. Core Globus services include: Globus Transfer for high-performance wide-area data movement using GridFTP over TLS; Globus Auth for federated identity and OAuth 2.0 authorisation, implementing the AARC Community AAI model described in Section 3.4; Globus Compute (formerly funcX) for function-as-a-service execution across distributed resources including HPC clusters and cloud VMs; and Globus Flows for orchestrating multi-step data management and computation workflows. Chard, Pruyne, McKee, Bryan, Raumann, Ananthakrishnan, Chard, and Foster (Chard et al., 2023), in

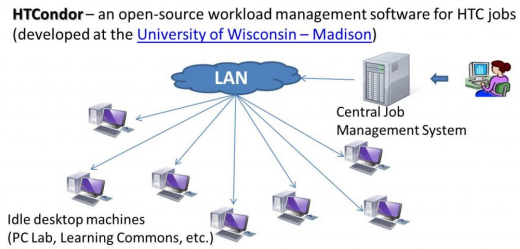
their peer-reviewed paper on Globus Automation Services published in Future Generation Computer Systems, document how Globus Flows enables the specification of research processes as reusable, cloud-hosted workflows spanning scientific instruments, data stores, and computing resources over timescales from seconds to weeks — precisely the multi-step, long-running workflows that the Collective layer of Section 1.2's hourglass architecture was designed for. Globus Transfer handles more than 1 exabyte of data transfers annually across the US research computing ecosystem and is the de facto standard for inter-site data movement (Chard et al., 2023).

EGI, DIRAC, and European Infrastructure

The European Grid Infrastructure (EGI) coordinates a federation of grid and cloud resources from more than 30 European countries. EGI's middleware strategy has shifted from the monolithic gLite stack toward a modular ecosystem of independently deployable components: the EGI DataHub for federated data access through OneData, OpenStack-based cloud IaaS through the OCCI API, and the EGI Workload Manager based on DIRAC. DIRAC (Distributed Infrastructure with Remote Agent Control), originally developed for the LHCb experiment at CERN around 2000, employs an agent-based architecture in which lightweight pilot agents deployed at each site actively pull work from central job queues rather than passively waiting for job submissions from the scheduler. This pull-based design provides exceptional resilience to site failures and network instability: if a site becomes unavailable, its pilot agents simply stop pulling work, and no jobs are lost or stranded at the inaccessible site. Stagni, Boyer, Tsaregorodtsev, Lytovchenko, Sailer, Haen, Burr, Bauer, Fayer, Martyniak, and Serfon (Stagni et al., 2024), in the most recent peer-reviewed account of DIRAC published in EPJ Web of Conferences at CHEP 2023, describe the ongoing transition of DIRAC toward its next-generation incarnation DiracX — a complete architectural rewrite using modern cloud-native technologies (FastAPI, OpenAPI, Elasticsearch, and OAuth 2.0 token-based authentication) while maintaining backward compatibility with existing DIRAC installations. This evolution directly mirrors the broader grid security transi-

tion documented in Section 3.4: DiracX's native token-based AAI replaces the legacy X.509 proxy certificate model that the original DIRAC inherited from the GSI. DIRAC currently manages workloads across more than 100 grid sites and has been adopted by communities in astrophysics, bioinformatics, and climate science beyond its origins in high-energy physics (Stagni et al., 2024).

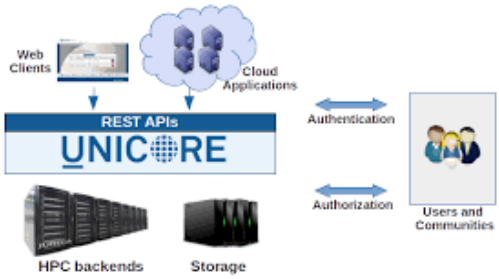
HTCondor and High-Throughput Computing



HTCondor, developed at the University of Wisconsin-Madison's Center for High Throughput Computing since 1984, is optimised for high-throughput computing (HTC) — executing very large numbers of independent jobs to maximise aggregate completion rate rather than minimising the latency of any individual job. HTCondor's ClassAd matchmaking language allows extraordinarily flexible expression of job requirements and resource capabilities, enabling automatic matching across arbitrary combinations of attributes including CPU architecture, memory, software availability, network connectivity, and user-defined site policies. Bockelman, Livny, Lin, and Prelz (Bockelman et al., 2020), in their peer-reviewed paper on the translational journey of the HTCondor-CE published in the Journal of Computational Science, document how the HTCondor Compute Entrypoint (CE) — the component that enables HTCondor's job management model to operate at grid site boundaries — evolved from an OSG-specific tool into a general-purpose grid gateway now deployed at sites across OSG, EGI, and WLCG. The paper's translational framing is instructive: it shows that HTCondor's architecture of decentralised matchmaking and flexible ClassAd policies is precisely what enables it to interoperate across the heterogeneous site policies and resource types that multi-domain grid environments contain. The Open Science Pool (OSPool), managed by the

OSG Consortium and based on HTCondor, provided more than 850 million CPU core-hours to US open-science researchers in 2023 — making it one of the most productive research computing infrastructures in the world (Graur et al., 2023).

UNICORE for HPC Integration



UNICORE (Uniform Interface to Computing Resources), developed in Germany and widely adopted across European HPC centres, provides a gateway-based integration approach particularly well suited to connecting traditional HPC batch computing environments to grid and cloud access layers. The UNICORE architecture centres on a layered gateway model in which a client-facing UNICORE/X gateway component translates job submission requests — expressed in the Job Submission Description Language (JSDL) or UNICORE's REST API — into the local resource manager's native format, whether SLURM, PBS, or LSF, insulating users and upper-level grid services from the heterogeneity of site-local schedulers. Benedyczak, Bała, and Streit (2016), in the peer-reviewed account of the UNICORE 7 architecture published in the proceedings of the 2011 IEEE/ACM International Conference on Grid Computing, document the layered UNICORE design and its integration with grid security and job management standards. Recent UNICORE versions have introduced REST API support alongside the existing SOAP-based interface, support for OIDC-based authentication through the AARC framework, and container job execution through Singularity/Apptainer — each of these additions directly implementing the architectural transitions described in Sections 3.4 and 3.5. UNICORE remains the primary grid-to-HPC integration middleware at major European HPC centres and is one of the reference access implementations recognised

by the EOSC Interoperability Framework (AARC – Authentication and Authorisation for Research and Collaboration, 2019).

Challenges, Case Studies, and Emerging Paradigms

Sections I, II, and III established the technical architecture of Distributed Grid Computing: the layered hourglass model and its service-oriented evolution (Section I); the scheduling, data locality, and workflow management mechanisms that orchestrate work across federated infrastructures (Section II); and the security infrastructure and middleware platforms that make cross-institutional trust and interoperability possible (Section III). Section IV now completes the analysis by examining four interconnected dimensions: the persistent and emerging challenges that the technologies of Sections I–III must address; concrete case studies demonstrating how production grid infrastructures have applied these technologies to some of the most significant scientific problems of the 21st century; and the convergence of grid, cloud, and edge computing paradigms that is defining the structural direction of the field. The challenges examined in Section 4.1 are not independent problems but consequences of the architectural and operational choices analysed in Sections I–III: the heterogeneity problem is the operational face of Section I's governance challenge; the scalability problem is the operational face of Section II's scheduling complexity; the sustainability problem connects directly to the energy cost of the data movement infrastructure described in Section 2.4. The case studies in Section 4.2 similarly demonstrate that the production realisations of grid computing — the WLCG, OSPool, ESGF, and GISAID — are not simply applications of the preceding sections' principles but stress tests that have driven the evolution of those principles, most directly in the HL-LHC computing challenge that motivated both the data locality innovations of Section 2.4 and the token-based security transition of Section 3.4. The convergence analysis of Section 4.3 then frames the forward trajectory of the field: hybrid grid-cloud architectures, edge computing integration, and AI-augmented scheduling represent not replacements for the architecture of Sections I–III but extensions of it, each preserving the federated governance model that makes grid computing

institutionally viable while substantially expanding the range of resources and workloads it can accommodate.

Current Challenges

Heterogeneity and Interoperability

The heterogeneity of grid environments — in hardware, software, middleware versions, and administrative policies — remains the most persistent operational challenge. A grid spanning dozens of sites must manage differences in CPU architectures, operating system versions, installed software libraries, local batch scheduler interfaces, storage system protocols, and network connectivity. Containerisation has emerged as one of the most effective responses to software heterogeneity. Kurtzer, Sochat, and Bauer (Kurtzer et al., 2017), in the peer-reviewed paper "Singularity: Scientific containers for mobility of compute" published in PLOS ONE, describe how the Singularity container runtime — now continued as the Apptainer project under the Linux Foundation — was designed specifically for HPC and grid environments with the requirement that containers must not require root privileges to execute. This design constraint, absent from Docker and other enterprise container runtimes, is what makes Singularity/Apptainer deployable in shared, multi-user grid environments without compromising site security policies. By packaging scientific applications with their complete software dependency stack — including libraries, compilers, and runtime environments — in a single portable image, Apptainer allows jobs submitted to heterogeneous grid sites to execute in reproducible, identical software environments regardless of what is installed on the host system. The adoption of containerised application delivery across OSG, EGI, and WLCG has significantly reduced the software deployment friction that previously complicated multi-site deployments, and is now considered a prerequisite capability for any production grid site.

Scalability and Information Management

As grid infrastructures scale to hundreds of sites and millions of jobs per day, the information management systems underpinning resource discovery and job monitoring face significant scalability challenges. Traditional hierarchical LDAP-based information systems — such as the original MDS (Monitoring and Discovery Service) in the Globus Toolkit described in Section 2.2 — struggle to maintain consistent, current information across hundreds of sites with thousands of compute nodes and dynamic resource availability. Propagation delays inherent in hierarchical information aggregation can result in scheduling decisions based on stale resource state information, leading to job failures or suboptimal placements — exactly the staleness problem that Section 2.2 identified as the fundamental limitation of polling-based information architectures. The EGI and WLCG have addressed this by deploying unified monitoring frameworks based on time-series databases and message brokers, as documented by Aimar, Aguado Corman, Andrade, Delgado Fernandez, Garrido Bear, Karavakis, Marek Kulikowski, and Magnoni (Aimar et al., 2019) for the WLCG MONIT infrastructure, providing near-real-time visibility at a scale not achievable with traditional polling approaches.

The scalability challenge has intensified with the onset of LHC Run 3 in 2022, which substantially increased monitoring data volumes across all WLCG experiments. Jashal, Kuznetsov, Legger, and Uzunoglu (2024), on behalf of the CMS Collaboration, in their CHEP 2023 paper published in EPJ Web of Conferences, document how CMS evolved its Run 3 monitoring infrastructure from bespoke custom-built applications toward a common open-source stack — built on ElasticSearch, VictoriaMetrics, Prometheus, InfluxDB, and Grafana — deployed on Kubernetes clusters to handle the heterogeneous monitoring data produced by workload management, data management, and submission infrastructure across approximately 150 sites worldwide. Kim and Bourilkov (Kim & Bourilkov, 2024), also in EPJ Web of Conferences at CHEP 2023, demonstrate how machine learning applied to the XRootD monitoring data harvested from the LHC computing infrastructure via the MONIT Elasticsearch database enables automatic detection of

underperforming or failing grid components — closing the loop between the near-real-time monitoring architecture of MONIT and the AI-enhanced operational intelligence discussed in Section 4.1.3. Together, these two 2024 papers confirm that scalable monitoring in production grid infrastructures is no longer just a data collection and display problem but an active site-reliability engineering challenge requiring the full DevOps monitoring stack and, increasingly, machine learning for automated anomaly detection at the scale of hundreds of sites and millions of daily jobs.

Artificial Intelligence Integration

The integration of AI and machine learning into grid computing operations is one of the most active research areas in 2024–2025, directly extending the algorithmic foundations laid in Section 2.3's examination of ML-enhanced scheduling. AI is being applied across four principal dimensions in contemporary grid environments. The first is predictive scheduling: using machine learning models trained on historical job traces to predict resource requirements and run times, enabling schedulers to make better-informed placement and backfill decisions. Zrigui, de Camargo, Legrand, and Trystram (Zrigui et al., 2022), in their peer-reviewed paper in the Journal of Parallel and Distributed Computing, demonstrate that online classification of jobs as "small" or "large" based on historical runtime data — rather than user-provided upper bounds, which are notoriously unreliable — significantly improves scheduling metrics including queue wait time and overall system utilisation, with the classifier adapting continuously to workload shifts. The second dimension is workflow-level deep reinforcement learning (DRL): training neural network policies to make sequential scheduling decisions over entire DAG-structured scientific workflows rather than individual jobs. Koslovski, et al. (Koslovski et al., 2024), in their peer-reviewed paper on DAG-based workflow scheduling using Actor-Critic Deep Reinforcement Learning published in Future Generation Computer Systems, show that DRL policies trained on synthetic and real workflow traces outperform classical heuristics (HEFT, list scheduling) on heterogeneous grids — confirming that the NP-completeness of multi-objective grid scheduling identified

by Hegde et al. (Hegde et al., 2024) in Section 2.1 is precisely the kind of intractable decision space where learned policies have a structural advantage over fixed heuristics. The third dimension is anomaly detection: using unsupervised learning to identify unusual monitoring patterns indicating resource failures or security incidents, as discussed in Section 4.1.2's account of Kim and Bourilkov (2024)'s ML-based monitoring of the WLCG infrastructure. The fourth, and most recent, dimension is LLM-assisted usability: translating natural language job specifications into formal job submission descriptions for non-specialist users. Adimora, Gundla, and Sun (Adimora et al., 2025), in their comprehensive survey of machine learning for HPC scheduling published in Cluster Computing, document all four dimensions and identify LLM-assisted job description translation as a particularly promising emerging application, while also identifying prompt injection as a new attack surface requiring careful mitigation in any deployment that exposes LLM interfaces to untrusted user input. The convergence of these four AI dimensions — predictive scheduling, reinforcement learning for workflow optimisation, anomaly detection, and LLM-assisted usability — suggests that the next generation of grid middleware will be characterised not by static heuristics but by continuously learning, adaptive operational intelligence trained on the rich job and monitoring telemetry that production grids generate at scale.

Data Management at Exascale

The volumes of scientific data that grid infrastructure must manage are growing at rates that challenge current storage, network, and software infrastructure simultaneously. The Square Kilometre Array (SKA) telescope, under construction in South Africa and Australia, represents the most demanding data challenge in the history of scientific computing. Scaife (Scaife, 2020), in the peer-reviewed paper "Big telescope, big data: towards exascale with the Square Kilometre Array" published in Philosophical Transactions of the Royal Society A, documents that each SKA telescope phase is expected to produce approximately 300 petabytes per year of observatory-level data products, resulting in an archive of approximately 8.5 exabytes over the 15-year lifespan of the initial high-

priority science programmes. Crucially, Scaife explains that the raw data rates from the SKA antennas — approximately 0.5–1 terabytes per second per telescope — are far too large to distribute: the SKA's Science Data Processors must perform in-situ calibration and imaging at each telescope site, reducing the data to manageable science products before distribution to a global network of SKA Regional Centres (SRCNet) for further analysis. The SRCNet architecture, endorsed by the SKAO governing council in March 2024, is explicitly modelled on the WLCG's tiered data distribution model described in Section 2.4, and adopts Rucio — the same distributed data management system used by WLCG — as its core data distribution layer, with Rucio Storage Elements at each Regional Centre node replicating observatory data products according to demand and network topology. Managing, distributing, and preserving this data will therefore require not just larger versions of existing infrastructure but qualitatively new approaches to computation-to-data architecture, federated storage governance, and metadata management. FAIR data management practices — now mandated by the European Commission and major US funding agencies — add explicit requirements for metadata cataloguing, persistent identifier assignment, and standardised data access (M. D. Wilkinson et al., 2016). The SKAO has explicitly committed to delivering SKA data under FAIR principles through the SRCNet, making it the first observatory-scale scientific data grid to embed FAIR compliance as a first-class design requirement from the outset rather than as a retrofit.

Environmental Sustainability

The environmental impact of large-scale computing infrastructure has become a significant concern as research organisations commit to carbon neutrality targets mandated by funding agencies and national governments. Britton, Campana, and Panzer-Stradel (Britton et al., 2024), in their peer-reviewed paper "A holistic study of the WLCG energy needs for the LHC scientific program" published in EPJ Web of Conferences at CHEP 2023, provide the most authoritative published estimate of the WLCG's energy consumption: the infrastructure is distributed across more than 170 data centres in over

40 countries, and its total energy consumption and projected evolution through the HL-LHC era depend on a complex interplay of luminosity, data volume, software efficiency, and the Power Usage Effectiveness (PUE) of individual data centre facilities. Britton et al. analyse how improvements in each of these dimensions — more efficient LHC experiment software, new hardware architectures, and modernised data centres — can substantially reduce the WLCG's energy footprint per unit of scientific output even as the absolute data volume grows. At the site level, the environmental cost of grid computing is highly geography-dependent. Researchers at Imperial College London (Imperial College London HEP Group, 2024). Grid infrastructure operators are increasingly adopting energy-aware scheduling strategies that direct workloads toward sites powered by renewable energy and schedule computationally intensive jobs during periods of low carbon-intensity electricity. ARM-based processor deployments at WLCG Tier-1 and Tier-2 sites — which the Glasgow and GridKa measurements confirm deliver approximately one-third lower energy consumption than equivalent x86 processors for the same HEP workloads — represent a hardware-level response to the same sustainability imperative. The convergence of energy-aware scheduling, renewable electricity procurement, and power-efficient hardware is now central to the WLCG's roadmap for delivering the HL-LHC scientific programme within the energy constraints that site operators and funding agencies are increasingly imposing.

Case Studies

The Worldwide LHC Computing Grid (WLCG) and LHC Run 3

The Worldwide LHC Computing Grid (WLCG) remains the world's largest scientific grid, linking more than 170 computing centres in 42 countries to process data from the Large Hadron Collider at CERN. The LHC's Run 3, which began in 2022 after a three-year shutdown for accelerator and detector upgrades, substantially increased data volumes. The ATLAS Collaboration (ATLAS Collaboration, 2025), in the comprehensive paper "Software and Computing for Run 3 of the ATLAS Experiment at the LHC" published in the European Physical Journal C, document that the upgraded LHC

generates raw data at rates that will reach 3 exabytes per year by the time of the High-Luminosity LHC (HL-LHC) upgrade planned for the early 2030s. The paper provides the most complete and up-to-date account of how the ATLAS experiment has evolved its entire software and computing stack — from detector simulation and event reconstruction through data management and distributed analysis — to operate efficiently across the heterogeneous WLCG infrastructure at Run 3 data rates, using containerised software environments, GPU-accelerated reconstruction, and data formats with up to 50-fold compression ratios compared to earlier generations. In response to these increased data volumes, the PanDA workload management system, coordinating ATLAS's use of WLCG resources and extending to cloud and HPC facilities, processed more than 2 billion computational tasks in 2023 across more than 180 computing sites worldwide (Maeno et al., 2024).

The Open Science Grid and Gravitational Wave Astronomy

The OSG Consortium's Open Science Pool (OSPool) has become one of the most productive research computing infrastructures for US open science. The LIGO-Virgo-KAGRA (LVK) collaboration uses OSPool resources extensively for the computationally intensive matched-filter analysis required to search for gravitational wave signals in detector data. The fourth observing run (O4), which began in 2023, is detecting gravitational wave events at an average rate of approximately one per week — a pace achievable only because of the distributed computing capacity provided by the OSPool and allied grid resources. Andrijauskas, Weitzel, and Wuerthwein (Andrijauskas et al., 2024), in their peer-reviewed paper on the Open Science Data Federation published in the ACM PEARC '24 proceedings, document OSPool operations including its scale of more than 850 million CPU core-hours provided to open science projects at zero marginal cost to individual research groups, demonstrating the economic value of the cooperative grid model. The LVK case is particularly instructive for grid computing: the computational bottleneck in gravitational wave detection is not the detectors themselves but the matched-filter analysis that must correlate gigabytes of detector data against

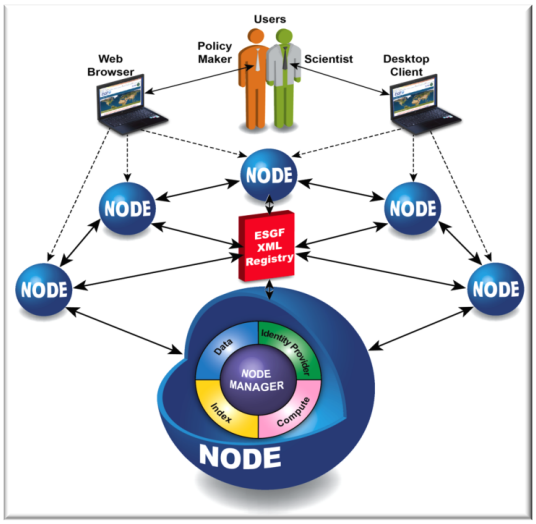
banks of hundreds of thousands of theoretical waveform templates — an embarrassingly parallel workload that maps almost perfectly onto the high-throughput computing model that OSPool and HTCondor were designed for, illustrating why the grid computing paradigm has proven so durable for data-intensive scientific discovery.

Grid Computing in Genomics: The COVID-19 Response

The COVID-19 pandemic generated unprecedented demand for large-scale genomic analysis, and the response demonstrated both the power and the limitations of distributed data grid infrastructure at global scale. The GISAID initiative — originally established as a global data grid for influenza genomic surveillance and documented by Shu and McCauley (Shu & McCauley, 2017) in Eurosurveillance — pivoted rapidly to become the primary repository for SARS-CoV-2 genome sequences, hosting more than 15 million viral genomes by 2023. GISAID's architecture — a federated database with data nodes at institutions in multiple countries, providing open access under a data use agreement that balances openness with attribution — is precisely the kind of distributed data grid infrastructure described in Sections 2.4 and 2.5: it provides the data locality, provenance tracking, and standardised access protocols that made continuously updated phylogenetic analyses of variant emergence possible across hundreds of research groups worldwide, as documented by Chen, Deng, Fang, Sun, Wu, Che, and colleagues (Chen et al., 2022) in their survey of the global landscape of SARS-CoV-2 genomic surveillance published in Nature Genetics. The Folding@home distributed computing project pivoted rapidly in March 2020 to focus on SARS-CoV-2 protein structure prediction. At its peak, Folding@home harnessed more than 2.4 exaFLOPS of computing power from more than 700,000 volunteer computers worldwide — briefly making it the most powerful distributed computing system ever assembled. Zimmerman, Porter, Ward, Singh, Vithani, Meller, and colleagues (Zimmerman et al., 2021), in their peer-reviewed paper published in Nature Chemistry, document how this distributed simulation capacity enabled the discovery of cryptic pockets across the SARS-CoV-2 proteome, including dramatic spike protein opening far beyond that observed experimentally, revealing new

targets for antiviral design that would have been computationally inaccessible without the exascale capacity that Folding@home's distributed architecture provided.

The Earth System Grid Federation



The Earth System Grid Federation (ESGF) is a peer-to-peer data grid that distributes climate model output from the Coupled Model Intercomparison Project (CMIP) to researchers worldwide. Petrie, Denvil, Ames, Levavasseur, Fiore, Allen, Antonio, Berger, Bretonnière, Cinquini, Dart, Dwarakanath, Druken, Evans, Franchistéguy, Gardoll, Gerbier, Greenslade, Hassell, Iwi, Juckes, Kindermann, Lacinski, Mirto, Nasser, Nassisi, Nienhouse, Nikonov, Nuzzo, Richards, Ridzwan, Rixen, Serradell, Snow, Stephens, Stockhause, Vahlenkamp, and Wagner (Petrie et al., 2021), in the definitive peer-reviewed account of the ESGF's CMIP6 data distribution operations published in Geoscientific Model Development, document that CMIP6 generated more than 20 petabytes of climate simulation data from over 100 global climate models at institutions in 30 countries, distributed through ESGF's network of primary data nodes in North America, Europe, and Asia-Pacific, serving more than 50,000 registered users. ESGF plays a direct role in the architectural principles examined throughout this paper: it implements the standardised data formats (NetCDF/CF conventions), standardised access protocols (OPeNDAP, GridFTP, HTTPS), federated metadata cataloguing, and persistent identifier assignment that the FAIR data management framework of Section 4.1.4 requires. ESGF played a

critical role in supporting the science underlying the IPCC Sixth Assessment Report (AR6) (Boehm et al., 2022). The ability to compare model output across dozens of independent modelling groups — made possible by ESGF's standardised data formats and access protocols — was essential to the AR6's characterisation of climate model uncertainty and the robustness of its projections.

Grid Computing Access in Kenya and Africa

In the African context, Distributed Grid Computing infrastructure has been developing along two complementary tracks: the construction of national research and education network (NREN) connectivity that provides the bandwidth prerequisite for grid participation, and the acquisition of grid credentials and access to international computing resources through that connectivity. Murumba and Micheni (Murumba & Micheni, 2017), in their peer-reviewed paper "Grid Computing for Collaborative Research Systems in Kenyan Universities" published in The International Journal of Engineering and Science, provide the most complete academic account of how KENET's infrastructure enables Kenyan university researchers to access international grid computing resources. They document that KENET digital certificates — issued by the KENET Certification Authority, which is accredited by EUGridPMA within the IGTF trust framework described in Section 3.2 — are accepted on both the Africa Grid and the European Grid (EGI), giving researchers at KENET member institutions access to enormous computing and storage resources that would otherwise be beyond the reach of any single Kenyan institution. The paper also documents KENET's 2016 training workshop on grid infrastructure and science gateways, held at the University of Nairobi and attended by more than 25 researchers and PhD students from Kenyan universities — confirming that the University of Nairobi scenario described in Section 3.1 of this paper is not hypothetical but reflects a documented, active participation pattern. Kenyan researchers in bioinformatics, epidemiology, and climate science access WLCG, OSG, and EGI grid resources through KENET's connectivity, enabling participation in international research collaborations — including the GISAID genomic surveillance described in Sec-

tion 4.2.3 — that would be impractical with local computing resources alone. The continued expansion of grid computing access to African research institutions — through KENET's IGTF-trusted credential infrastructure, UbuntuNet Alliance's regional backbone, and the AfricaConnect EU-funded projects — represents an important and ongoing dimension of the global democratisation of scientific computing.

The Convergence of Grid, Cloud, and Edge Computing

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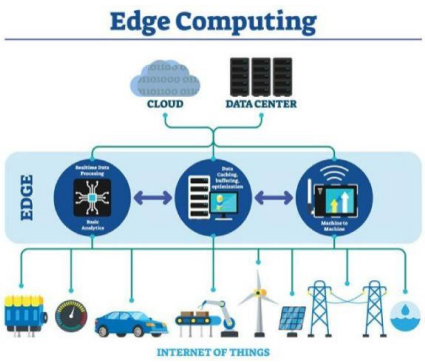
Hybrid Grid-Cloud Architectures

The relationship between grid computing and cloud computing has evolved from perceived competition to productive complementarity, driven by the recognition that dedicated grid infrastructure and commercial cloud resources each have structural advantages that the other cannot replicate. Dedicated grid infrastructure provides predictable, long-term capacity under institutional control, deep integration with community-specific data and software environments, and data sovereignty — the guarantee that sensitive research data does not leave institutional-controlled storage. Commercial cloud resources from providers such as AWS, Google Cloud, and Azure provide elastic scaling for peak demand without permanent infrastructure investment, access to specialist hardware (high-memory instances, FPGA accelerators, GPU clusters) that no single institution can afford to maintain continuously, and global geographic distribution for workloads requiring low-latency access to diverse data sources.

Contemporary research computing environments deploy hybrid architectures that combine both. The NSF's ACCESS programme, documented by Boerner, Deems, Furlani, Knuth, and Towns (Boerner et al., 2023) in their PEARC '23 paper, formally embodies this hybrid model: ACCESS provides a coordinated portfolio of national HPC resources alongside cloud allocations and science gateways, with allocation and access managed through a common framework — precisely the federated resource management architecture described in Section 2.1. Wu, Tang, Yu, Yang, and Sun (Wu et al., 2025), in their

survey of task scheduling in geo-distributed computing published in IEEE Transactions on Parallel and Distributed Systems, document how cloud bursting — the dynamic overflow of grid workloads to commercial cloud capacity during demand peaks — has become a standard operational strategy for WLCG Tier-1 and Tier-2 sites, with the principal technical challenge being the minimisation of data movement costs when cloud instances must access data stored in on-premises grid storage systems. The Globus platform provides this capability through Globus Compute and Globus Transfer services that abstract over the heterogeneity of underlying computing and storage environments — whether traditional grid batch systems or cloud virtual machines — behind consistent REST APIs (Chard et al., 2023).

Edge Computing and Real-Time Scientific Grids



The growing volume of data generated by scientific instruments at remote field sites has motivated the integration of edge computing into grid architectures. Rather than transmitting all raw sensor data to centralised data centres, edge computing deploys processing capacity at or near the data source, performing initial data reduction before transmitting significantly compressed data products to the central grid. The SKA telescope is the most dramatic example. Scaife (Scaife, 2020), in the authoritative account of SKA computing published in Philosophical Transactions of the Royal Society A, documents that each SKA telescope is expected to produce approximately 300 petabytes per year of science-ready data products after on-site processing — data rates that are only achievable because the SKA's Central Signal Processor performs real-

time correlation and calibration at the telescope site, reducing terabytes-per-second raw antenna streams to manageable science data products before distribution through the global SRCNet data grid.

The SKAO website confirms that the two SKA Science Data Processors will each operate at approximately 130–135 petaFLOPS — comparable to the current top-ranked supercomputers on the TOP500 list — making the SKA's on-site edge processing infrastructure one of the most computationally intensive scientific edge deployments ever attempted. This edge-grid architecture directly implements the data locality principles analysed in Section 2.4: computation is performed as close to the raw data as physically possible, with only reduced, science-ready products distributed to the global analysis grid — the same principle that motivates Rucio's replica-aware scheduling in the WLCG, applied at the extreme of the data-volume spectrum. Designing, building, and operating this global data grid infrastructure for the SKA is one of the most technically challenging distributed systems engineering problems currently under active development.

AI and Machine Learning Workloads on Grid Infrastructure

The rapid growth of AI and machine learning as tools for scientific data analysis has created new demands on grid infrastructure that extend the scheduling and data management challenges examined in Sections 2.1–2.3. Training large deep learning models requires GPU computing resources and the ability to distribute training across multiple nodes — a workload pattern characterised by tightly coupled, communication-intensive parallel tasks that is fundamentally different from the embarrassingly parallel, loosely coupled job streams for which traditional grid schedulers were optimised. This distinction matters architecturally: the HTCondor and PanDA scheduling models described in Section 3.6 were designed for workloads where jobs are largely independent; distributed deep learning frameworks such as Horovod and PyTorch Distributed require instead that co-scheduled jobs communicate continuously during training, making gang scheduling and network topology awareness first-class scheduler requirements.

Grid middleware platforms are being extended to support GPU-enabled job submission, checkpoint-based fault tolerance for long-running training jobs, and these tightly-coupled distributed training frameworks, as documented by Adimora, Gundla, and Sun (Adimora et al., 2025) in their comprehensive survey of machine learning for HPC and grid scheduling published in Cluster Computing. The EGI Notebooks service and PANGEO cloud-native geoscience analytics platform represent emerging models for making grid and cloud resources accessible for AI/ML scientific workflows through familiar Jupyter notebook interfaces, combining the data access capabilities of distributed data grids with the interactive computing environment of cloud platforms. Calatrava and colleagues (Calatrava et al., 2023), in their survey of EOSC services for multidisciplinary scientific applications published in Computer Science Review, document how the EOSC's federated resource portfolio — encompassing cloud, HPC, and data services — is being progressively extended to support AI-intensive scientific workflows, with the EGI Notebooks and PANGEO integrations representing the user-facing layer of a multi-tier infrastructure that descends through the AARC-federated identity framework of Section 3.4 to the distributed storage and compute resources of the WLCG and EGI.

Conclusion

This paper has demonstrated that Distributed Grid Computing is not a legacy paradigm but the mature federated infrastructure that currently underpins the most consequential scientific discoveries of the 21st century. Section I established that the five-layer hourglass architecture — standardising minimally at the interface between resource providers and consumers — has governed grid computing for three decades, surviving every transition from Globus to OGSA to contemporary EOSC-scale deployments (Foster et al., 2001); (Grigoryeva & Klimentov, 2024). Section II showed that grid scheduling is NP-complete in its general form, driving adoption of pilot-job architectures, the Rucio–PanDA data locality stack, and AI-enhanced approaches including deep reinforcement learning (Hegde et al., 2024); (Maeno et al., 2024). Section III

traced the security evolution from X.509 proxy delegation to OAuth 2.0 token-based federated identity, documenting a transition driven by operational scale rather than cryptographic failure, alongside parallel modernisation of Globus, DIRAC, HTCondor, and UNICORE (Dack et al., 2024). Section IV confirmed through five production case studies — the WLCG, OSPool, GISAID, ESGF, and KENET — that grid infrastructure is delivering science at scales that no single institution could achieve alone, while the convergence with cloud and edge computing is extending, not replacing, the federated model (ATLAS Collaboration, 2025); (Petrie et al., 2021); (Murumba & Micheni, 2017).

Research communities that fail to invest in federated grid infrastructure face a concrete penalty: the scientific workloads documented in this paper — LHC collision analysis, matched-filter gravitational wave searches, global genomic surveillance, climate model intercomparison — become either computationally infeasible or captive to commercial cloud providers who cannot guarantee the data sovereignty and community governance that large scientific collaborations require. Institutions in emerging research economies, where NREN connectivity and IGTF-accredited credentials are the difference between participation and exclusion, face this penalty most acutely (Murumba & Micheni, 2017). On 14 September 2015, two black holes merging 1.3 billion light-years away produced a signal lasting less than a fifth of a second — and detecting it required more than 100 institutions operating as one federated grid. Distributed Grid Computing is not merely one approach among many: it is the infrastructure through which humanity currently conducts its most ambitious science, and it will remain so for as long as the most consequential scientific questions demand more than any one institution can provide.

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